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ORIGINAL PAPER

Spatiotemporal analysis of drought and rainfall in Pakistan via Standardized Precipitation Index: homogeneous regions, trend, wavelet, and influence of El Niño-southern oscillation

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Abstract

The phenomenon of drought is common in the world, especially in Pakistan. El Niño-Southern Oscillation (ENSO) influences the spatial and temporal variability of drought and rainfall in Pakistan. Therefore, the objectives of this study are to identify homogeneous rainfall regions and their trend regions, as well as the impact of ENSO phases. In this study, monthly rainfall data from 44 weather stations are used during 1980–2019. Moreover, descriptive and exploratory statistics tests (e.g., Pettitt and Mann-Kendall—MK), Sen method, and cluster analysis (CA) are evaluated along with the annual Standardized Precipitation Index (SPI) on spatiotemporal scales. ENSO occurrences were classified based on the Oceanic Nino Index (ONI) for region 3.4. Using the cophenetic correlation coefficient (CCC) and a significance level of 5%, seven methods were applied to the rainfall series, with the complete method (CCC > 0.9082) being the best. According to the CA method, Pakistan has four groups of homogeneous rainfall (G_1, G_2, G_3, and G_4). Descriptive and exploratory statistics showed that G_1 differs from the other groups in size and spatial distribution. Pettitt's technique identified the most extreme El Niño years in terms of spatial and temporal drought variability, along with the wettest months (March, August, September, June, and December) in Pakistan. Non-significant increases in Pakistan's annual precipitation were identified via the MK test, with exceptions in the southern and northern regions, respectively. No significant increase in rainfall in Pakistan was found using the Sen method, especially in regions G_2, G_3, and G_4. The severity of the drought in Pakistan is intensified by El Niño events, which demand attention from public managers in the management of water resources, agriculture, and the country's economy.

1 Introduction

Drought is a natural phenomenon defined as water scarcity in a certain period of time, varies between months to a year, or even several years (McKee et al. 1993; Gonçalves et al. 2021), being common across the world (WMO 2012). Numerous studies are based on quantifying the severity degree via drought indexes and different calculation methods prevailing in the literature, for example, the Palmer Drought Severity Index (PDSI), Rainfall Anomaly Index (RAI), Vegetation Condition Index (VCI), Soil Moisture Deficit Index (SMDI), Standardizerd Runoff Index (SRI), and Single

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Z-Index (Heim Jr. 2002; Oliveira Júnior et al. 2012; Beguería et al. 2010; Sobral et al. 2018; Fung et al. 2020; Costa et al. 2021; Gonçalves et al. 2021). These indices identify the duration, frequency, and intensity of the drought (Kim et al. 2014; Carmo and Lima 2020; Ho et al. 2021; Oliveira-Júnior et al. 2021). Furthermore, they use hydro-meteorlogical variables such as rainfall, evapotranspiration (ET), surface air temperature, soil moisture, and runoff as input data (Mishra and Singh 2010; Zargar et al. 2011; Beguería et al. 2010; Sobral et al. 2018).

The Standardized Precipitation Index (SPI) uses solely rain as input for the most often used index in the scientific literature for drought identification (McKee et al. 1993), which is a consensus among several experts (Mishra and Singh 2010; Zargar et al. 2011; Oliveira Júnior et al. 2012; Khan and Gadiwala 2013; Kim et al. 2014; Lyra et al. 2017; Terassi et al. 2020; Oliveira-Júnior et al. 2021; Alsubih et al. 2021). The World Meteorological Organization (WMO) (WMO 2012) has recognized the SPI as a reliable tool for monitoring meteorological droughts.

Adnan et al. (2018) previously studied 15 drought indices in Pakistan and compared them using various statistical tests, concluding that the SPI, Standardized Precipitation Evapotranspiration Index (SPEI), and Reconnaissance Drought Index (RDI) had a solid capacity to monitor drought in the country. Throughout Pakistan, the severity and frequency of SPEI-based droughts increase during the main seasons, Rabbi and Kharif, Rabi (November to April) and Kharif (May to October) and two critical periods for crop growth in term of water supply, namely critical Rabi (March–April) and critical Kharif (May) (Mohsenipour et al. 2018). Drought intensity increases primarily in dry and semi-arid regions for both crop growing seasons (Ahmed et al. 2018).

According to the findings by Ahmed et al. (2018), droughts have been severe and recurrent in Pakistan, particularly in arid and semi-arid regions (Salma et al. 2012; Anjum et al. 2012), increasing the social vulnerability of the country's population (Ahmed et al. 2018; Adnan et al. 2018) by undermining the country's livelihood, agriculture, and economy (Hussain and Lee 2009; Hanif et al. 2013; Ullah et al. 2021). Ullah et al. (2021) have examined meteorological drought in multiscales (3-, 6-, and 12-month intervals) during a 35-year period using SPI and SPEI. Rather than using reanalysis products, the authors employed in situ data (Climatic Research Unit - CRU TS, National Centers for Environmental Prediction version II - NCEP-2, European Center for Medium-Range Weather Forecasts Version-5 - ERA-5, and Modern-Era Retrospective analysis for Research and Applications version II - MERRA-2). The findings revealed that the products (i.e., CRU TS and MERRA-2) could accurately identify drought in the country's south, except for overestimation in the east and west sectors (ERA-5 and NCEP-2). Hanif et al. (2013) investigated the processes and precursors of drought dynamics in Pakistan, finding that exceptional drought episodes in the Pacific and Indian oceans are significantly associated with wind patterns and the intrinsic weather system.

In Pakistan, several studies on drought assessment and monitoring have been conducted (e.g., Salma et al. 2012; Anjum et al. 2012; Adnan et al. 2018; Ullah et al. 2021). However, few studies have correlated with the phases of climate variability mode, such as ENSO (Ahmad et al. 2010; Ahmad et al. 2014; Hina et al. 2021), followed by the definition of homogeneous rainfall regions (Hussain and Lee 2009) in the country and the best cluster analysis (CA) method, while still using a consistent time series observed (Anjum et al. 2012; Salma et al. 2012; Khan and Gadiwala 2013; Xie et al. 2013; ADB 2017; Mallick et al. 2022). Therefore, motivated by the gaps that still exist in the meteorological trends and phenomena, the objectives of the study are as follows: (i) to assess spatiotemporality of the drought based on the annual SPI; (ii) identify homogeneous regions and rainfall trends; and (iii) assess the influence of ENSO phases on the dynamics of drought in Pakistan.

2 Materials and methods

2.1 Study area

Pakistan is geographically, situated between 24–37° N latitudes and 62–75° E longitudes in the western zone of south Asia, has five Provinces, namely Punjab, Sind, Balochistan, Gilgit-Baltistan, and Khyber Pukhtunkhwa (KP).

Furthermore, Pakistan is separated between tropical and subtropical zones, with 75% of the overall landmass (mostly in Pakistan's South) receiving less rainfall in the winter and experiencing high temperatures in the summer (Adnan 2010).

2.2 Rainfall data

Monthly rainfall data from 44 weather stations were used in the study, as indicated in Table 1. The Mangla station, with a gap of 21 years in the time series, was eliminated from the analyses. The monthly rainfall data of the time series comprise the period 1980–2019.

2.3 Pettitt and Mann-Kendall tests

The current research used the Pettitt test (Pettitt 1979) to identify the years of possible abrupt changes in Pakistan's

Table 1 Geographical coordinates (latitude and longitude in (°)), altitude (m), mean annual rainfall (mm year^{-1}) and identifiers (ID) of Pakistan's 44 weather stations in the period 1980–2019

ID	Stations	Latitude (°) (N)	Longitude (°) (E)	Altitude (m)	Average annual rainfall (mm year^{-1})
1	Astore	35.35°	74.86°	2374	474.53
2	Badin	24.30°	68.54°	3	227.02
3	Bahawalpur	29.35°	71.69°	118	173.25
4	**Balakot**	34.54°	73.35°	989	**1514.89**
5	Barkhan	29.89°	69.52°	1104	423.52
6	Bunji	35.64°	74.63°	1455	164.15
7	Chhor	25.51°	69.78°	6	238.33
8	Chillas	35.42°	74.09°	1168	200.36
9	Chitral	35.76°	71.77°	1421	453.67
10	Dalbandin	28.88°	64.39°	848	82.05
11	Dera Ismail Khan	31.86°	70.90°	178	321.52
12	**Dir**	35.19°	71.87°	1343	**1424.76**
13	Drosh	35.56°	71.80°	1324	554.91
14	Faisalabad	31.45°	73.13°	189	391.53
15	Gilgit	35.88°	74.46°	1441	141.58
16	Gupis	36.22°	73.44°	2209	202.56
17	Hyderabad	25.39°	68.35°	26	155.57
18	**Islamabad**	33.68°	73.04°	526	**1250.51**
19	Jacobabad	28.24°	68.38°	55	134.08
20	Jhelum	32.94°	73.72°	232	889.30
21	Jiwani	25.05°	61.77°	19	98.51
22	Karachi (Airport)	24.90°	67.16°	22	173.41
23	Khanpur (PBO)	28.63°	70.65°	87	143.21
24	**Kotli**	33.50°	73.90°	598	**1219.99**
25	Lahore(PBO)	31.52°	74.35°	215	691.12
26	Lasbella	25.87°	66.71°	136	165.20
27	Moen-Jo-Daro	27.32°	68.13°	52	95.69
28	Multan (PBO)	30.15°	71.52°	122	213.90
29	**Murree**	33.90°	73.39°	2178	**1738.67**
30	**Muzaffarabad Airport**	34.34°	73.50°	815	**1484.51**
31	Nawabshah	26.24°	68.39°	29	153.22
32	Nokkundi	28.82°	62.75°	682	35.43
33	Padidan	26.77°	68.29°	42	122.84
34	Panjgur	26.97°	64.08°	971	91.09
35	**Parachinar**	33.90°	70.08°	1738	**1044.70**
36	Pasni	25.25°	63.41°	11	102.34
37	Peshawar Airport	33.98°	71.51°	360	506.43
38	Quetta	30.17°	66.97°	1641	263.57
39	Rohri	27.66°	68.89°	65	107.45
40	Sargodha	32.07°	72.68°	188	498.27
41	**Sialkot**	32.49°	74.52°	252	**1005.16**
42	Sibbi	29.55°	67.88°	137	181.17
43	Skardu (PBO)	35.32°	75.55°	2227	235.02
44	Zhob (PBO)	31.34°	69.46°	1436	281.85

average rainfall time series. The Pettitt test is a non-parametric test, which checks whether two samples X_1, X_2 …, X_t and X_{t+1}, X_{t+2} …, X_T belong to the same population. The statistics of $k(t) = U_{t,T}$ performs a count of a member of the 1st sample to be greater than the member of the 2nd sample. According to below Eq. (1):

$$U_{t,T} = \sum_{i=1}^{t} \sum_{j=t+1}^{T} \operatorname{sgn}\left(X_j - X_i\right) \quad (1)$$

where sgn is the coefficient given by the following conditions: sgn = +1 if $(X_j - X_i) > 0$; sgn = 0 if $(X_j - X_i) = 0$; sgn = −1 if $(X_j - X_i) < 0$; T is the size of time series; and i and j denote the time indices associated with individual values.

The statistics $U_{t,T}$ is then calculated for the values of $1 \leq t < T$, and so the $k(t)$ statistic of the test corresponds to the maximum in the absolute value of $U_{t,T}$ and, thus, the years where the abrupt change occurs are estimated, given by Eq. (2):

$$k(t) = {}_{1 \leq t < T}{}^{\text{Max}} \left|U_{t,T}\right| \quad (2)$$

The Pettitt test locates the point where there is an abrupt change in the mean of a time series, and its significance is given by Eq. (3):

$$p \cong 2 \exp\left[\frac{-6k(t)^2}{\left(T^3 + T^2\right)}\right] \quad (3)$$

The sudden change point is t, where the maximum of $k(t)$ occurred, then the critical values of k (k_{crit}) are calculated by Eq. (4):

$$k_{\text{crit}} = \pm\sqrt{\frac{\ln\left(\frac{p}{2}\right)\left(T^3 + T^2\right)}{6}} \quad (4)$$

The Mann-Kendall (MK) test (Mann 1945; Kendall 1975) was applied to Pakistan's rainfall time series, where x_i is of n terms ($1 \leq i \leq n$) and the MK test statistic is set to $x = x_1, x_i, x_j..., x_n$ according to Eq. (5). This test is also known as a distribution-free (non-parametric) test.

$$S = \sum_{j=i+1}^{n} \operatorname{sgn}\left(x_j - x_i\right) \quad (5)$$

where x_i is the estimated data of the sequence of values, n is the length of the time series, and sgn = 1 for $(x_j - x_i)>0$; sgn = 0 for $(x_j - x_i)=0$; sgn = −1 for $(x_j - x_i)<0$.

For time series $x_1, x_2, x_3..., x_n$ with a high number of terms ($n>4$) under the null hypothesis (H_0) of an absence of a trend, S will present a normal distribution with mean and variance, where the variance of S is denoted by Var(S) from Eq. (6):

$$Var(S) = \frac{n(n-1)(2n+5)}{18} \quad (6)$$

In this way, with the data repetitions, the variance is given by Eq. (7):

$$Var(S) = \frac{1}{18}\left[n(n-1)(2n+5) - \sum_{p=1}^{g} t_p\left(t_p - 1\right)\left(2t_{p+5}\right)\right] \quad (7)$$

where n is the number of observations; t_p is the number of observations with equal values in a specific p_{th} group; and g is the number of groups containing equal values in the data series in a particular p_{th} group. The second term represents an adjustment for censored data.

By testing the statistical significance of S for H_0 based on a two-tailed test, which in turn can be rejected for high values of the Z statistic given by Eq. (8):

$$Z_{\text{MK}} = \begin{Bmatrix} \frac{S-1}{\sqrt{\text{Var}(S)}} \text{ for } S > 0 \\ 0 \text{ for } S = 0 \\ \frac{S+1}{\sqrt{\text{Var}(S)}} \text{ for } S < 0 \end{Bmatrix} \quad (8)$$

Based on the Z_{MK} statistic (Table 2), the decision is made to accept or reject H_0 that is, H_0 is accepted when the time series has no trend and rejected in favor of the alternative hypothesis (H_1) when there is a trend in the time series, in short:

H_0: There is no statistically significant trend in the time series; that is, the trend is statistically insignificant for p-value>α;
H_1: There is a statistically significant trend in the time series for p-value>α.

For the analysis of the results in Pakistan, the Z-statistic values via the SPI-12 map are used. Positive values ($Z > 0$) show an increasing trend, while negative values ($Z < 0$) show a decreasing trend. The significance level (α) adopted in the study was $\alpha = 0.05$ for the MK and Pettitt tests. If the probability p of the MK test is less than the level α, $p < \alpha$, there is a significant trend, while a value of $p > \alpha$, an insignificant trend. When there is no trend, the value of Z is close to 0, and the value of p is close to α.

Table 2 Classification of the Z_{MK} trend in the confidence interval from −1.96 to +1.96

Category	Scales
Significant increasing trend - SIT	$Z_{\text{MK}} > +1.96$
Non-significant increased trend - NSIT	$Z_{\text{MK}} < +1.96$
No trend - NT	$Z_{\text{MK}} = 0$
Non-significant reduction trend - NSRT	$Z_{\text{MK}} > -1.96$
Significant reduction trend - SRT	$Z_{\text{MK}} < -1.96$

2.4 Standardized Precipitation Index

The Standardized Precipitation Index (SPI) formulation is based on the density and Gamma probability function (Eq. (5)) that is calculated every month, where α is the shape parameter ($\alpha > 0$), β is the scale parameter (β>0) which is determined using the maximum likelihood method; x is the amount of rainfall that can vary according to α and β. The assigned values are normalized and transformed to a normal distribution (i.e., mean zero and variance one) (McKee et al. 1993; Oliveira Júnior et al. 2012). The SPI calculation uses only rainfall values (Lyra et al. 2017; Carmo and Lima 2020) and categorizes a drought event either when the SPI presents continuous negative values or when the SPI is positive, and the drought event ends (McKee et al. 1993; Oliveira Júnior et al. 2012). Further details on the mathematical formulations and the statistical procedures used in calculating the SPI can be found in McKee et al. (1993), being given by Eq. (9):

$$g(x) = \frac{1}{\beta^{\alpha}\Gamma(\alpha)} x^{\alpha-1} e^{-\frac{x}{\beta}} \tag{9}$$

where $\Gamma(\alpha)$ is the gamma function.

After the calculation, the SPI was categorized according to Table 3 and analyzed on the annual scale (SPI-12). For the analysis of the SPI, the "SCI" package from the software library R version 4.1.1 (R Core Team 2020) was used.

2.5 Multivariate analysis

The statistical technique of cluster analysis (CA) was applied to the pluviometric monthly temporal series (definition of homogeneous regions) and SPI-12 in Pakistan. This study applies the complete agglomerative hierarchical method to the 44 weather stations in the study area in a dendrogram (Wilks 2011). Moreover, the measure of dissimilarity is also calculated in this study using the square of the Euclidean distance (Eq. (10)) (Lyra et al. 2014; Brito et al. 2017; Jardim et al. 2021).

Table 3 Classification of the Standardized Precipitation Index (SPI), according to McKee et al. (1993)

SPI	Classification
≥ 2.00	Extremely wet
1.00 to 1.99	Very wet
0.50 to 0.99	Moderately wet
0.49 to −0.49	Close to normal
−0.50 to −0.99	Moderately dry
−1.00 to −1.99	Very dry
≤ −2.00	Extremely dry

$$d_e = \left[\sum_{j=1}^{n}\left(P_{p,j} - P_{k,j}\right)^2\right]^{0.5} \tag{10}$$

where d_e is the Euclidean distance, and $P_{p,j}$ and $P_{k,j}$ are the quantitative variables j of individuals p and k, respectively.

We assessed the degree of adjustment of average linkage by utilizing the cophenetic correlation coefficient (CCC). This coefficient measures the association between the dissimilarity matrix (phenetic matrix F) and the matrix resulting from the simplification provided by the grouping method (cophenetic matrix c). The CCC is based on the Pearson coefficient (r), calculated between the dissimilarity matrix and the matrix resulting from the grouping (Sokal and Rohlf 1962). Thus, the higher the r-value is, the less is distortion. Similarly, the dendrograms with CCC < 0.7 indicate the inadequacy of the CA technique (Sokal and Rohlf 1962; Oliveira-Júnior et al. 2017). The CCC is defined by Eq. (11).

$$CCC = r_{coph} = \frac{\sum_{i=1}^{n-1}\sum_{j>i}^{n}\left(c_{ij} - \bar{c}\right)\left(d_{ij} - \bar{d}\right)}{\sqrt{\sum_{i=1}^{n-1}\sum_{j>i}^{n}\left(c_{ij} - \bar{c}\right)^2}\sqrt{\sum_{i=1}^{n-1}\sum_{j>i}^{n}\left(d_{ij} - \bar{d}\right)^2}} \tag{11}$$

$$\bar{c} = \frac{2}{n(n-1)}\sum_{i=1}^{n-1}\sum_{j>i}^{n} c_{ij} \tag{12}$$

$$\bar{d} = \frac{2}{n(n-1)}\sum_{i=1}^{n-1}\sum_{j>i}^{n} d_{ij} \tag{13}$$

where CCC is the cophenetic correlation coefficient, c_{ij} and d_{ij} are elements of the ith row and jth column of the cophenetic and the original distance matrix, respectively, and n is the number of elements. Moreover, the $\bar{c}$ and $\bar{d}$ are the arithmetic averages of c_{ij} and d_{ij}, respectively, as in Eqs. (12) and (13).

2.6 ENSO

The Oceanic Nino Index (ONI), which is used to identify El Niño (warm phase), La Niña (cold phase), and ENSO Neutral events in the Tropical Pacific Ocean, was employed in the study (Huang et al. 2017). The ONI must have a value equal to or greater than the threshold for at least 3 consecutive 3-month periods of sea surface temperature (SST) exceeding 0.5°C for each occurrence classified as weak, moderate, strong, or very strong El Niño (NOAA/CPC 2021). Only the SPI-12 is used in this study to detect drought episodes the annual scale in the time series, as well as their association with ENSO phases, which may be accessed at the following website: https://ggweather.com/enso/oni.htm.

2.7 Continuous wavelet transformation

The periodicity of the drought cycle in Pakistan was evaluated using SPI time series. The SPI-12 was calculated from monthly rainfall data, then accumulated annual rainfall. Using the gamma distribution, rainfall is simply transformed into a standard normal variable (Yerdelen et al. 2021). In this study, the variability in the data sets is identified using continuous wavelet transformation, from Eq. (14):

$$wf\Big)(z,a) = \frac{1}{\sqrt{z}} \int_{-\infty}^{+\infty} f(x)\varphi * \left[\frac{x-a}{z}\right] dx \tag{14}$$

Here, a is the location parameter, z is scaling factor, ψ^* corresponds to the complex conjugate of cwf and $f(x)$ refers to SPI time series. Wavelet scale changes with time and the translation factor gives rise to scalogram. When the predefined value is crossed by wavelet coefficients, the anomalous pattern is observed (Hafeez et al. 2022). Ideal threshold was retrieved by iterative process. The Morlet wavelet was implemented for decomposition of the SPI data series because it has the ability to decompose hydrological series in time and frequency (Rashid et al. 2015).

To study the variant pattern of rainfall in Pakistan on a time scale from 1980 to 2019 using wavelet transformation

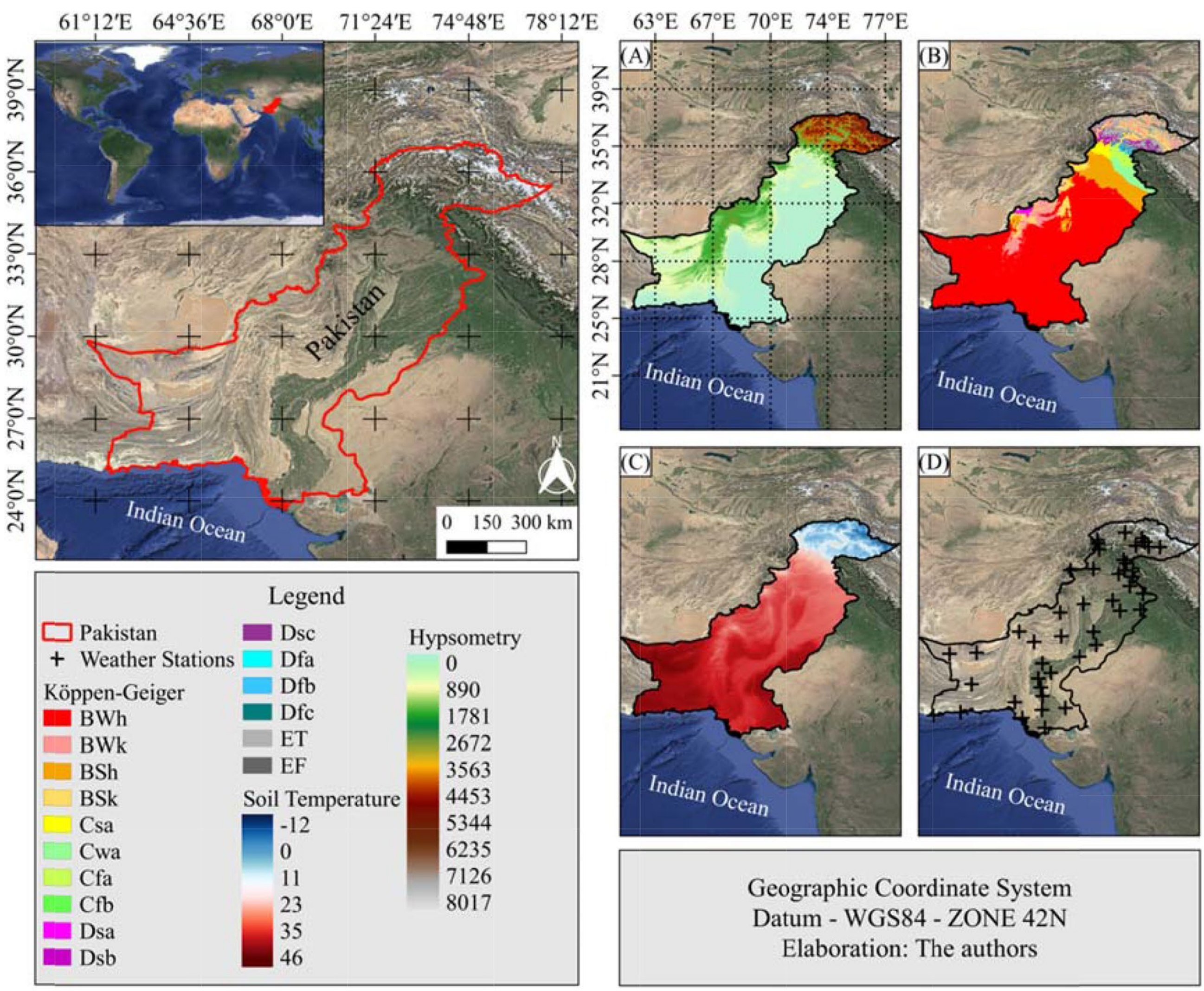

Fig. 1 Map of study area location and hypsometry of Pakistan (**A**), climate classification on the basis of de Köppen-Geiger (**B**), temperature in in (°C) via MODIS product (SRTM - Shuttle Radar Topography Mission, 30 m) (**C**) the location map of the weather stations (+) (**D**)

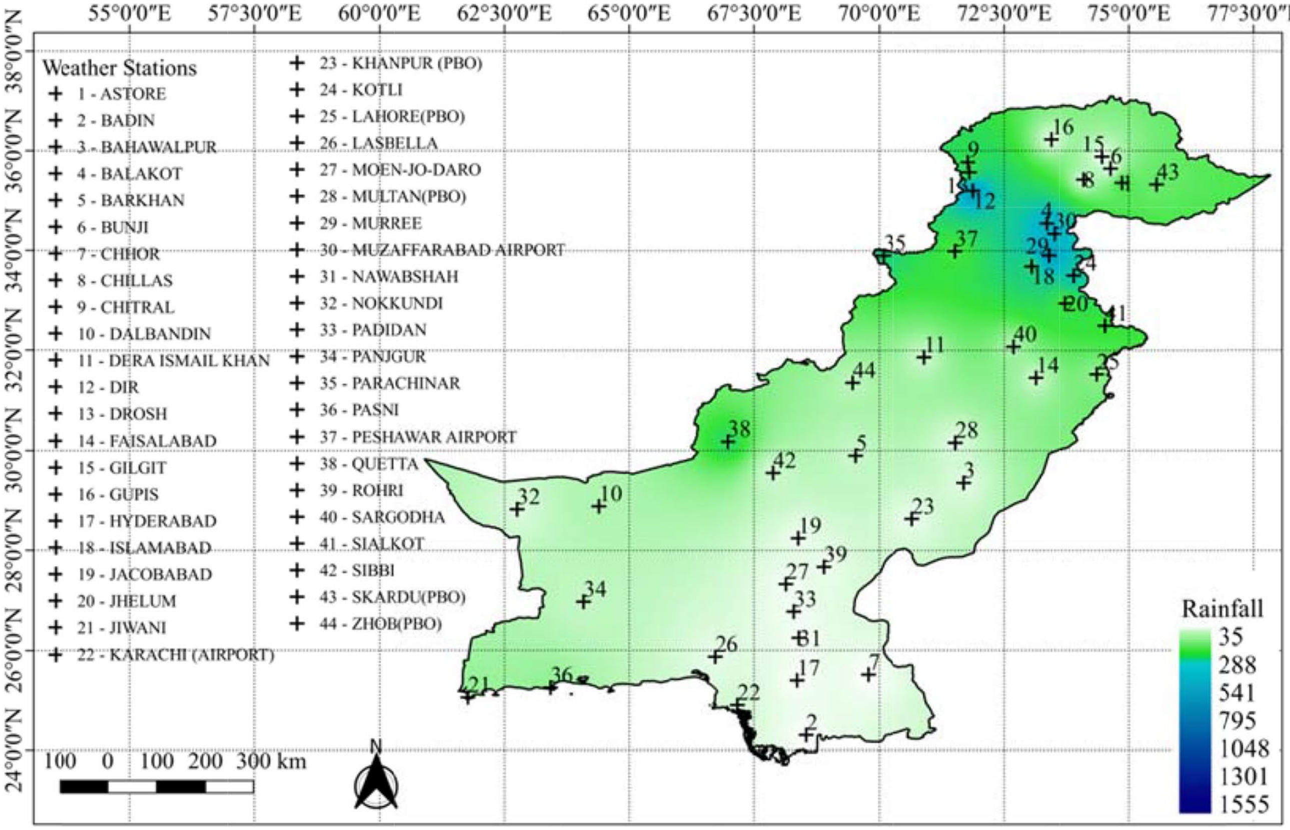

Fig. 2 Spatial distribution of mean annual rainfall (mm year^{-1}) in Pakistan during 1980–2019

in different regions, the data is divided into four groups. The continuous wavelet transformation (CWT) convert 1-Dimensional signal which is time series data into time and frequency domains in 2-Dimensional space for the purpose of estimating the dominating and evident periodicity and turning points in SPI data series.

3 Results and discussion

3.1 Climatology and rainfall seasonality

In the 40 years of data, Pakistan's average annual rainfall demonstrated the establishment of a rainfall gradient from the South to the North, with variations ranging from 38.21 to 1596.00 mm (Fig. 2). The Indian Ocean coastline influences the South, due to lower elevation values. The Balochistan Plateau influences the North, Indus Plain, Hindu Kush, and Himalaya Mountains influence the North (Fig. 1), with values exceeding 5000 m (Hussain and Lee 2009). According to Hussain and Lee (2009), the plains of Punjab and Sindh are responsible for about 60% of all rainfall in Pakistan during the monsoon season. Monsoon rains are the most critical hydrometeorological resource as they help agriculture or cause flooding in the country (Eckstein et al. 2020; Hina et al. 2021).

This gradient causes variation in average annual rainfall, owing to the fact that Pakistan receives less rain than other South Asian nations (e.g., Bangladesh, Bhutan, India, Maldives, Nepal, and Sri Lanka) (Hussain and Lee 2009; Adnan et al. 2018; Hina et al. 2021). However, some stations in Pakistan stand out, such as Balakot (ID 4), Dir (ID 12), and Islamabad (ID 18)—the country's capital; Kotli (ID 24); Murree (ID 29); Muzaffarabad Airport (ID 30); Parachinar (ID 35), and Sialkot (ID 41), all of which have annual rainfall records over 1000 mm (Table 1).

All the stations mentioned above are located in the north of the country, that is, in climatic zones A and B, as previously defined by Salma et al. (2012) and Hanif et al. (2013). Both authors found that the annual accumulated rainfall increased in the high latitudes of the country, contrary to the results obtained in this study (Fig. 2).

Monthly rainfall in Pakistan fluctuates due to the rainfall gradient between South and North, which is connected with the monsoon wind regime and the disturbances in Western Pakistan (Salma et al. 2012), although the distribution of monthly rainfall in Pakistan is not uniform. Still, the rain

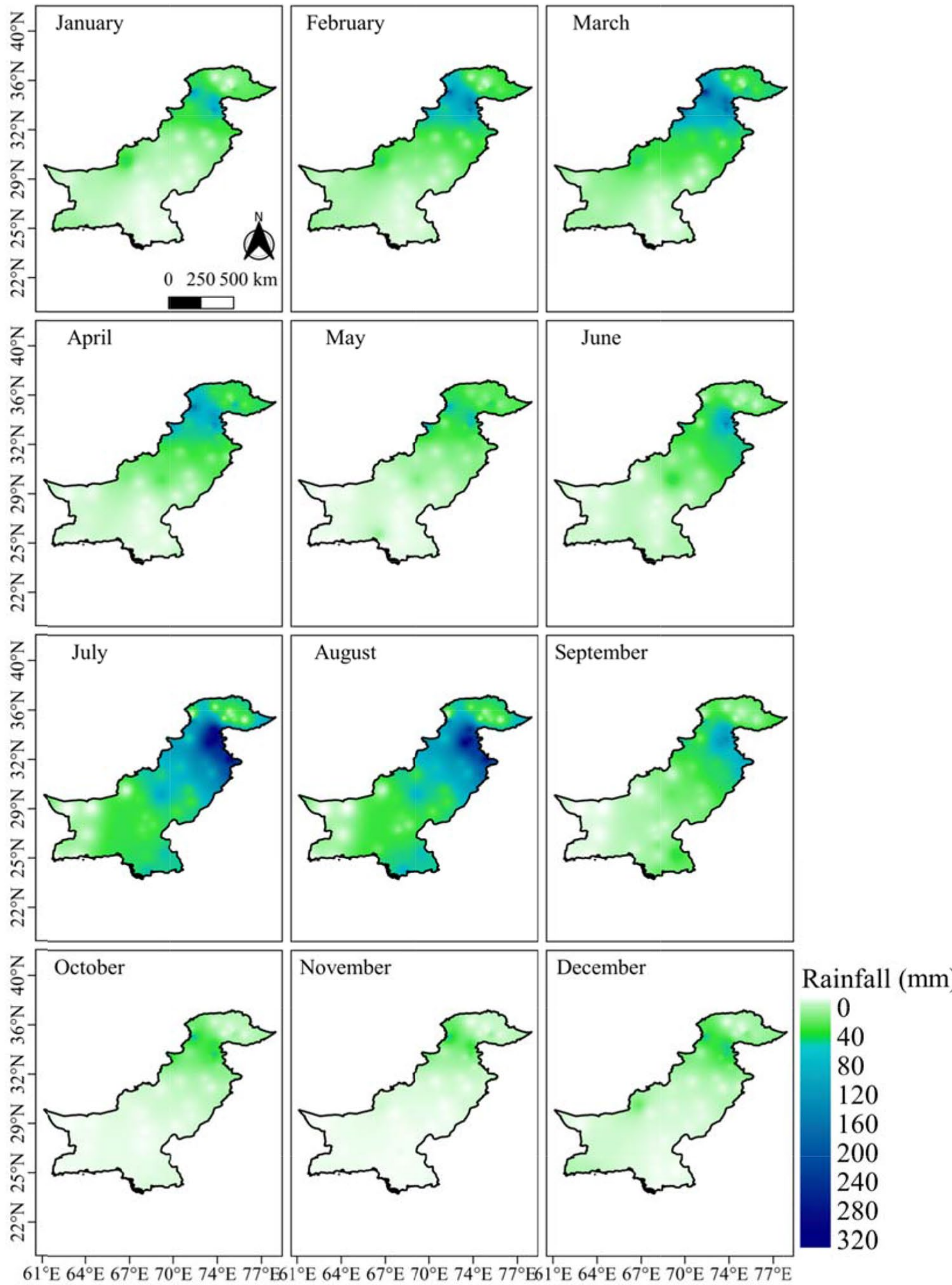

Fig. 3 Spatial distribution of mean monthly rainfall (mm month^{-1}) in Pakistan during 1980–2019

does not fall all year long (Fig. 3). During the months of July to September, Pakistan experiences monsoon rains that affect the country's eastern and northeastern regions. In the North and Northeastern regions of the country, there is a substantial amount of rainfall (Hussain and Lee 2009; Adnan et al. 2018; Hina et al. 2021). Notably, heavy monsoon-related rains, followed by flash floods and rivers, caused disasters in considerable part of Pakistan in 2010 (Sayama et al. 2012). From December to March, heavy rain fall on the region due to western disturbances from Iran and Afghanistan (Salma et al. 2012), which are then followed by the effect of Iran's Northern Mountains and the Balochistan Plateau. The wettest months are October to March (Hussain and Lee 2009).

3.2 Homogeneous rainfall regions

The yearly rainfall series in Pakistan were clustered using seven different approaches in this study. The procedures were assessed using the CCC, which was subjected to a 5% significance level, as indicated in Table 4. Because the complete approach was significant (*S*) and CCC > 0.9082, it stood out among the other techniques. Average, McQuity,

Table 4 Cophenetic correlation coefficient (CCC) and significance, respectively, were used to the temporal rainfall series (1980–2016) using cluster analysis method

ID	†Methods	CCC	‡Significance
1	Ward	0.6750	NS
2	Single	0.8255	NS
3	Complete	0.9082	S
4	Average	0.9039	S
5	McQuitty	0.9076	S
6	Median	0.9005	NS
7	Centroid	0.9055	S

†Linkage methods

‡NS, S is not significant and significant, respectively, with significance testing at 0.05 probability level

and centroid were three other approaches that stood out in the analysis, all of which had CCC > 0.90 and were all significant. The other approaches were ineffective, as the dendrogram was insufficient with a CCC of 0.70 (Sokal and Rohlf 1962; Wilks 2011; Oliveira-Junior et al. 2017) and was not significant (NS).Ward's method is widely used in the scientific literature (Modarres and Sarhadi 2011; Brito et al. 2017; Oliveira-Júnior et al. 2017; Silva et al. 2021) and presented the worst results for Pakistan.

The sum of the squares of the clusters for the 44 evaluated stations (Fig. 4a) allowed to identify an ideal number of four homogeneous groups in Pakistan. Unlike Mallick et al. (2022) based on a 40-year time series of rainfall from 22 weather stations in Saudi Arabia, where K-means clustering and hierarchical clustering were used in principal component analysis (PCA), which defined three homogeneous groups. The dendrogram is shown in Fig. 4b, with a percentage of 24.97% for the fenon line (cut line with the Euclidean distance) (Wilks 2011; Brito et al. 2017) and the homogeneous pluviometric groups formed. Group G_1 (229.65 ± 109.48 mm) is the largest group formed with 34 stations, followed by group G_2 (1346.63 ± 285.98 mm) with four stations and groups G_3 (861.86 ± 225.37 mm) and G_4 (1171 ± 359.22 mm), with three stations each (Table 5).

The G_1 group, despite the greater number of stations, presented the lowest values of median (209.29 mm), standard deviation (SD), and mean ($\overline{x}$). Unlike the other groups, with emphasis on groups G_2 and G_4 with the highest values of median, SD and $\overline{x}$, being such stations belong to the region with the highest rainfall records (Fig. 2). The maximum, minimum, and amplitude (AMP) values of the annual rainfall stood out in the G_2 group, with the minimum in the Dir station (ID 36 = 911.40 mm year^{-1}), the maximum in the Murree station (ID 38 = 2433.80 mm year^{-1}), and AMP = 1246.50 mm year^{-1} (ID 35 = Balakot), followed by the G_4 group, with minimum at Islamabad station (ID 42= 607.00 mm year $^{-1}$), maximum at Parachinar station (ID 44 = 2356.80 mm year^{-1}), and AMP = 1826.90 mm year $^{-1}$ (ID 44). As for the exceptions, they were in the G_3 group, with the minimum at the Lahore station (PBO) (ID 40 = 333.70 mm year^{-1}), the maximum at the Sialkot station (ID 38 = 1642.80 mm year $^{-1}$), and AMP = 1642.00 mm year^{-1} (ID 41). Highlight again for the G_1 group, with the lowest values between minimum (0 mm year^{-1}), maximum (949.80 mm year^{-1}) and AMP (187.60 mm year^{-1}) (Table 5) .

Homogeneous groups were used for all of the coefficients of variation (CVs, %) below 100%. Thus, except for two stations (ID 24 = 108.52% and ID 25 = 102.49%), all-time series data in group G_1 were homogeneous (Wilks 2011; Gois et al. 2020). In contrast to the other groups, the G_1 group had a larger prevalence of negative lower bounds in the exploratory statistics of yearly rainfall data at 21 stations. For G_2 and G_3 groups, the Murree (2501.34 mm year^{-1}) and Balakot (746.55 mm year^{-1}) stations were the upper and lower limits, respectively, while the Sialkot (1710.48 mm year^{-1}) and Lahore stations were the G3 and G4 groups' lower and higher limits, respectively. However, there was a discrepancy between the maximum interquartile range (IQR) found in groups G_2 (ID 24), G_3 (ID 3), and G_4 (237.98 mm year^{-1}, ID 28) (Table 5). The non-parametric statistical tests of MK and Pettitt are the most critical instruments in time series analysis (Gois et al. 2020). The years 1998 (El Niño powerful) were found using the Pettitt test, indicating a 17-fold occurrence (Table 5), 1991 (El Niño Strong), 2004, and 2005 (El Niño Strong) being the most significant occurrences in the time series. The years 2010 (La Nia Strong) and 2014 (El Niño Weak) were the exceptions (NOAA/CPC 2021), in contrast to Hina et al. (2021), who identified 1987, 1991, 2000, 2001, 2002, and 2005 from the crossover of information between the SPI and the Single *Z*-Index. The El Niño of the Century (Slingo and Annamalai 2000) occurred during the 1997–1998 cycle, and this El Niño, which was in the extreme category, caused a 5-year drought throughout Southwest Asia, including Pakistan (Ahmad et al. 2014, 2010; ADB 2017; Hina et al. 2021). The Pettitt test identified the months with the ,most significant sudden changes in monthly rainfall in Pakistan, being March (7 changes), August and September (6 changes), and June and December (4 changes) (Table 5). It is noteworthy that June to September is considered the monsoon season in Pakistan, which contributes almost 57% of the total annual rainfall (Hussain and Lee 2009).

The spatial distribution of homogenous groups revealed that the G_1 group in southern and northern Pakistan has higher stations, from the coast to the region with the most remarkable hypsometry totals (Fig. 5). The G_2 group lies near the plateau and plains, while the G_3 and G_4 groups are towards the mountain range's base. The transitional rainfall regime in Pakistan is characterized by the least rainy group

(G_1) between the coast and the mountain range and the wettest groups (G_2, G_3, and G_4) in the north and windward parts of the mountain range.

As an example, in the south, north, and northwest, where the highest mountains in the world (peak K2-8200 m) are located, homogeneous groupings are dispersed according to the country's physiography. Pakistan's Indus plains, Punjab and Sindh, are considered to be central, while the deserts of the southeast and the dry lands of southwest are considered to be peripheral (Chaudhry and Rasul 2000). Droughty conditions are found in 60% of the country according to the Köppen-Geiger classification "BWh" (Fig. 1). The climate is characterized by a hot, dry summer and a cold, snowy winter in the Himalayas. In the central region, semi-arid to dry conditions prevail, whereas coastal areas have their own distinct climate (Karim et al. 2020).

3.3 SPI annual

Spatial The prevalence of the very and extremely dry categories was seen in the spatial distribution of SPI-12 in the years identified by the Pettitt test. In the years 2005 and 2010, there are exceptions. In 2005, the category near to normal predominated in the eastern part of Pakistan. Still, in 2010, it prevailed in the western part of the nation, with occurrences of the same and highly humid categories in the country's extreme north, where the highest elevations are found (Fig. 1). In the years 1991, 2004, and 2014, severely dry conditions were observed in the regions of Balochistan,

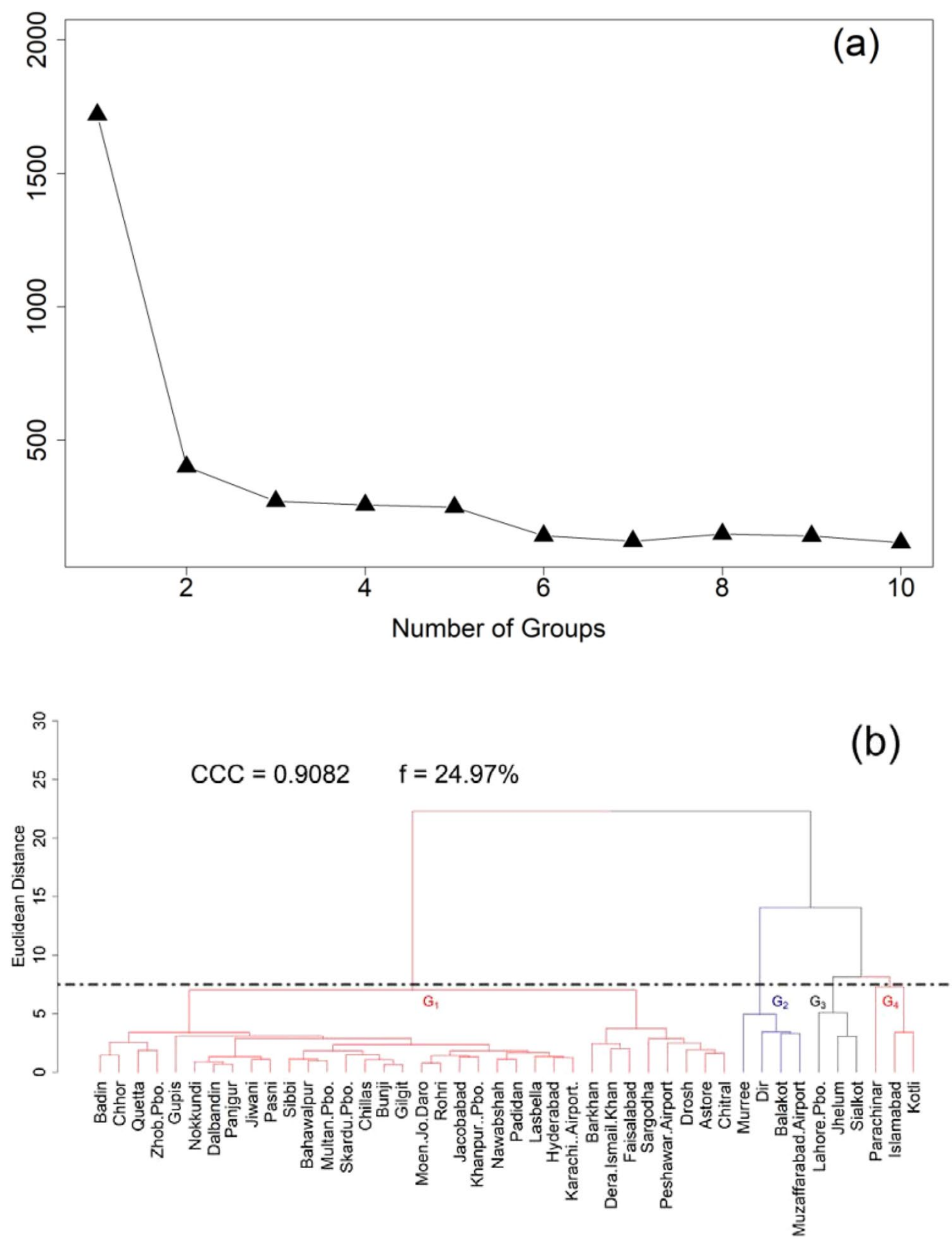

Fig. 4 **a** Sum of squares of the groups, and **b** average connection in the form of a dendrogram for homogeneous groups of rainfall (G_1, G_2, G_3, and G_4) for the 44 weather stations in Pakistan during 1980–2019

Table 5 The Pettitt and Mann-Kendall (MK) tests of the 44 weather stations belonging to homogenous groups in Pakistan between 1980 and 2019 have been summarized

Groups	ID	Stations	$\bar{x}$	Md	SD	Values			MK test	
						Minimum	Maximum	AMP	τ	Sen
			mm year^{-1}							
G_1	1	Astore	474.53	453.95	128.61	240.20	873.70	633.50	−0.0291	−0.0072
	2	Badin	227.02	182.10	173.14	0.00	913.90	913.90	0.0133	0.0000
	3	Bahawalpur	173.25	147.70	91.62	41.90	511.30	469.40	0.0114	0.0000
	4	Barkhan	423.52	392.75	148.58	98.10	784.00	685.90	0.0008	0.0000
	5	Bunji	164.15	154.50	62.05	61.80	339.40	277.60	0.0571	0.0024
	6	Chhor	238.33	206.60	152.35	4.60	550.50	545.90	0.0148	0.0000
	7	Chillas	200.36	184.65	87.05	84.80	569.40	484.60	−0.0158	0.0000
	8	Chitral	453.67	456.70	116.24	214.40	728.90	514.50	−0.0305	−0.0038
	9	Dalbandin	82.05	72.15	51.49	3.50	203.70	200.20	−0.1015	0.0000
	10	Dera IsMayl Khan	321.52	295.85	129.27	123.40	756.30	632.90	0.0047	0.0000
	11	Drosh	554.91	574.80	129.23	289.70	867.20	577.50	−0.0525	−0.0156
	12	Faisalabad	391.53	373.40	133.09	172.40	806.70	634.30	0.0412	0.0010
	13	Gilgit	141.58	135.20	45.93	64.40	267.50	203.10	0.0722	0.0041
	14	Gupis	202.56	148.10	155.08	5.30	675.60	670.30	0.1611	0.0096
	15	Hyderabad	155.57	121.40	128.91	8.50	524.90	516.40	0.0059	0.0000
	16	Jacobabad	134.08	94.45	115.59	13.90	485.80	471.90	0.0411	0.0000
	17	Jiwani	98.51	67.40	91.06	0.00	386.40	386.40	−0.1176	0.0000
	18	Karachi (Airport)	173.41	157.30	124.48	0.00	481.50	481.50	0.0072	0.0000
	19	Khanpur (PBO)	143.21	111.00	103.62	8.60	409.00	400.40	0.0938	0.0000
	20	Lasbella	166.04	143.05	103.31	8.70	474.60	465.90	0.1420	0.0000
	21	Moen-Jo-Daro	95.69	79.05	87.70	0.00	413.10	413.10	0.0933	0.0000
	22	Multan(Pbo)	213.90	215.30	88.89	59.60	513.20	453.60	0.0500	0.0000
	23	Nawabshah	153.22	110.75	145.01	0.00	637.30	637.30	0.0505	0.0000
	24	Nokkundi	35.43	20.05	38.45	0.00	187.60	187.60	−0.0166	0.0000
	25	Padidan	122.84	83.00	125.90	3.30	546.90	543.60	−0.0154	0.0000
	26	Panjgur	91.09	79.85	57.66	9.00	304.30	295.30	−0.0530	0.0000
	27	Pasni	102.34	83.60	81.73	0.00	356.20	356.20	−0.0207	0.0000
	28	Peshawar Airport	506.43	463.95	164.03	252.30	904.50	652.20	0.0556	0.0144
	29	Quetta	263.57	243.50	155.89	93.50	949.80	856.30	−0.0506	0.0000
	30	Rohri	107.45	88.50	87.85	5.80	452.30	446.50	0.0448	0.0000
	31	Sargodha	498.27	493.45	140.63	266.00	767.30	501.30	0.0414	0.0070
	32	Sibbi	181.17	172.35	90.90	9.70	371.90	362.20	0.0022	0.0000
	33	Skardu (PBO)	235.02	226.50	90.62	93.30	495.40	402.10	0.0016	0.0000
	34	Zhob (PBO)	281.85	282.90	96.24	109.60	495.00	385.40	−0.0386	0.0000
Total			**229.65**	**209.29**	**109.48**					
G_2	35	Balakot	1299.60	1668.30	309.53	1069.50	2316.00	1246.50	−0.0524	−0.0463
	36	Dir	1256.53	1582.08	260.17	911.40	2149.00	1237.60	−0.0624	−0.0517
	37	Murree	1507.53	1905.05	315.86	1264.40	2433.80	1169.40	−0.0507	−0.0510
	38	Muzaffarabad Airport	1322.85	1675.35	258.35	970.10	2153.20	1183.10	−0.0184	−0.0154
Total			**1346.63**	**1707.69**	**285.98**					
G_3	39	Jhelum	889.30	907.25	189.52	532.60	1335.70	803.10	−0.0123	−0.0032
	40	Lahore (PBO)	691.12	637.20	205.28	333.70	1232.50	898.80	−0.0182	−0.0016
	41	Sialkot	1005.16	936.30	281.30	544.50	1642.00	1097.50	−0.0214	−0.0063
Total			**861.86**	**826.92**	**225.37**					
G_4	42	Islamabad	1250.51	1203.30	326.79	607.00	1900.00	1293.00	−0.0273	−0.0136
	43	Kotli	1219.99	1220.00	272.33	707.80	1789.80	1082.00	−0.0216	−0.0134
	44	Parachinar	1044.70	895.75	478.54	529.90	2356.80	1826.90	0.1665	0.1104

Table 5 (continued)

Total			**1171.73**	**1106.35**	**359.22**					
Groups	**ID**	**Stations**		**Limits**		**Quartiles**			**Pettitt's test**	
			CV	L_i	L_u	Q_1	Q_3	**IQR**	**Years**	**Months**
			%	**mm year**$^{-1}$						
G_1	1	Astore	27.10	159.38	761.58	385.20	535.75	150.55	2005	May
	2	Badin	76.27	−160.06	618.04	131.73	326.25	194.53	1985	Sep
	3	Bahawalpur	40.36	−28.36	343.34	111.03	203.95	92.93	2006	Feb
	4	Barkhan	35.08	70.93	754.53	327.28	498.18	170.90	1998	Sep
	5	Bunji	37.80	17.01	313.71	128.28	202.45	74.18	1993	Apr
	6	Chhor	63.93	−107.79	531.91	132.10	292.03	159.93	1991	Nov
	7	Chillas	43.45	26.85	351.85	148.73	229.98	81.25	1985	Nov
	8	Chitral	25.62	189.94	721.04	389.10	521.88	132.78	2010	Sep
	9	Dalbandin	62.76	−68.73	224.48	41.23	114.53	73.30	1998	Apr
	10	Dera IsMayl Khan	40.21	−33.68	688.33	237.08	417.58	180.50	1989	Nov
	11	Drosh	23.29	238.74	861.84	472.40	628.18	155.78	1998	Jun
	12	Faisalabad	33.99	143.06	636.36	328.05	451.38	123.33	2004	Mar
	13	Gilgit	32.44	30.11	236.21	107.40	158.93	51.53	2003	Jan
	14	Gupis	76.56	−141.86	509.04	102.23	264.95	162.73	1991	Dec
	15	Hyderabad	82.87	−155.06	418.24	59.93	203.25	143.33	2004	Jul
	16	Jacobabad	24.36	−179.29	410.61	41.93	189.40	147.48	1988	Jun
	17	Jiwani	92.44	−126.71	293.59	30.90	135.98	105.08	1998	Feb
	18	Karachi (Airport)	71.79	−210.44	559.46	78.28	270.75	192.48	1994	Jun
	19	Khanpur (PBO)	72.35	−103.41	359.69	70.25	186.03	115.78	2006	Feb
	20	Lasbella	62.22	−73.94	384.36	97.93	212.50	114.58	2005	Mar
	21	Moen-Jo-Daro	91.65	−121.50	284.50	30.75	132.25	101.50	1991	Dec
	22	Multan (PBO)	41.56	−26.49	438.81	148.00	264.33	116.33	2004	Mar
	23	Nawabshah	30.56	−145.39	390.91	55.73	189.80	134.08	2004	Jul
	24	Nokkundi	108.52	−50.76	109.54	9.35	49.43	40.08	1988	Apr
	25	Padidan	102.49	−174.66	387.44	36.13	176.65	140.53	1998	Mar
	26	Panjgur	63.30	−28.99	205.31	58.88	117.45	58.58	1998	Mar
	27	Pasni	79.86	−163.73	362.88	33.75	165.40	131.65	2014	Mar
	28	Peshawar Airport	32.39	24.31	976.21	381.28	619.25	237.98	1991	Dec
	29	Quetta	59.14	−56.25	528.75	163.13	309.38	146.25	1998	Jun
	30	Rohri	81.76	−90.94	285.76	50.33	144.50	94.18	2004	Sep
	31	Sargodha	28.22	54.69	920.99	379.55	596.13	216.58	2001	Mar
	32	Sibbi	50.17	−74.06	436.84	117.53	245.25	127.73	1998	Aug
	33	Skardu (PBO)	38.56	−36.66	491.24	161.30	293.28	131.98	1986	Jan
	34	Zhob (PBO)	34.15	−39.90	595.90	198.53	357.48	158.95	1998	Aug
G_2	35	Balakot	20.43	746.55	2221.35	1299.60	1668.30	368.70	1998	Aug
	36	Dir	18.26	768.20	2070.40	1256.53	1582.08	325.55	1996	Aug
	37	Murree	22.17	911.24	2501.34	1507.53	1905.05	397.53	1998	Sep
	38	Muzaffarabad Airport	17.40	794.10	2204.10	1322.85	1675.35	352.50	1998	Aug
G_3	39	Jhelum	21.31	377.10	1365.90	747.90	995.10	247.20	1998	Oct
	40	Lahore (PBO)	29.70	109.60	1261.40	541.53	829.48	287.95	1998	Oct
	41	Sialkot	40.70	300.88	1710.48	829.48	1181.88	352.40	1998	Oct
G_4	42	Islamabad	26.13	191.09	2360.19	1004.50	1546.78	542.28	1998	Sep
	43	Kotli	22.32	298.73	2145.33	991.20	1452.85	461.65	1998	Aug
	44	Parachinar	39.23	309.68	1480.48	748.73	1041.43	292.70	2005	Dec

$\bar{x}$ the average, *Md* median, *SD* standard deviation, *AMP* amplitude, τ Kendall tau, *Sen* Sen curvature method, *CV* coefficient of variation, L_i inferior limit, L_u is upper limit, *Q1*, *Q3* the lower and upper quartiles, *IQR* the interquartile range

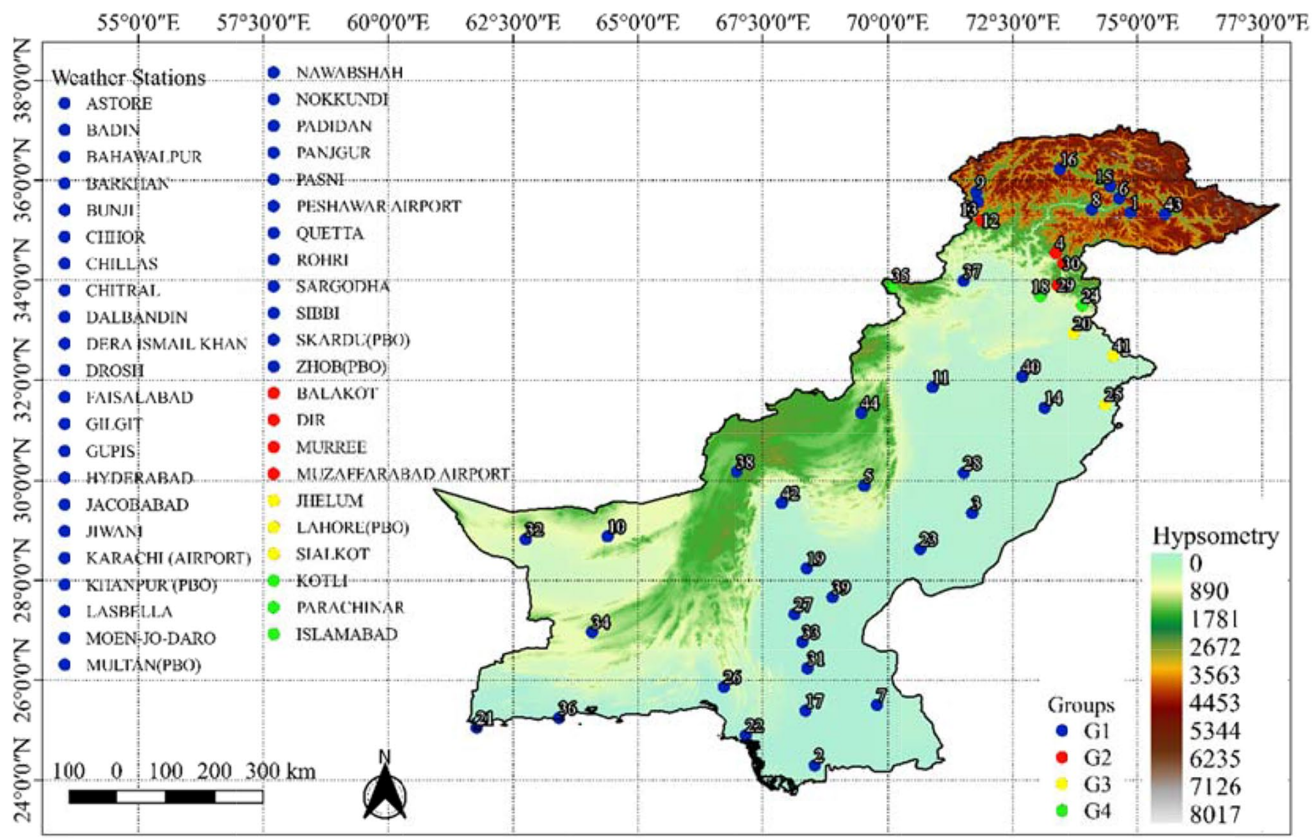

Fig. 5 Pakistan's spatial distribution of homogeneous rainfall groups (G_1, G_2, G_3, and G_4) and its hypsometry (m)

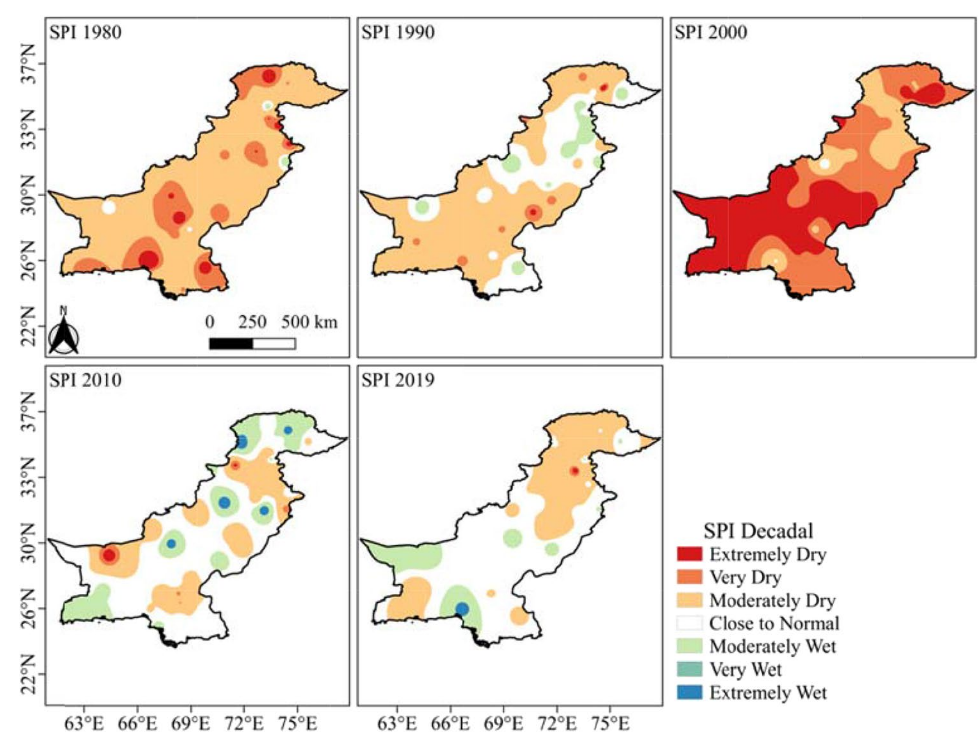

Fig. 6 According to the Pettitt test, the spatial distribution of SPI-12 in Pakistan in the years 1991, 1998, 2004, 2005, 2010, and 2014

Sindh, and Punjab, as opposed to highly wet conditions in particular Pakistani regions in the years 2005 and 2010 (Fig. 1).

According to ONI (Huang et al. 2017; NOAA/CPC 2021), the years 1991 (El Niño strong), 1998 (El Niño very strong), the 2004/2005 cycle (transition between El Niño weak and La Niña weak), 2010 (El Niño moderate), and 2014 (El Niño moderate) had an impact on the dynamics of drought in Pakistan. The main advantage of the SPI is that it can be calculated in a temporal multiscale (Mckee et al. 1993; Mishra and Singh 2010; Kim et al. 2014; Alsubih et al. 2021) and can be used to assess the impacts of rainfall deficits as well as to compare different locations (Carmo and Lima 2020; Costa et al. 2021; Gonçalves et al. 2021). Pakistan is divided into tropical and subtropical zones, with arid regions accounting for 75% of the total landmass (mainly in the South) and making it more vulnerable to drought (Adnan 2010) (Fig. 6).

El Niño events were analyzed using the Pettitt test, and the geographical analysis based on SPI-12 was examined for the years of very strong El Niño events (Huang et al. 2017; NOAA/CPC 2021) (Fig. 7). According to the SPI in Table 3, in Pakistan in the 1982/83 cycle, there were higher records of the categories very dry and close to normal. There were a few exceptions to this rule in 1982, when exceptionally humid and dry conditions were prevalent in the north of the country. It is worth noting that in 1983, the extremes of Pakistan saw a particularly dry period, while the middle of the country experienced a period close to usual. For the first time, data from an Asian precipitation grid (APHRODITE - *Asian Precipitation Highly-Resolved Observational Data Integration Towards Evaluation*) was used to calculate the SPI, and the PCA (principal component analysis) was used to identify droughts in Pakistan's central and southern regions. Between 1998 and 2002, Pakistan had a severe drought as a result of the 1997/98 El Niño (Slingo and Annamalai 2000; ADB 2017; Ahmad et al. 2010; Ahmad et al. 2014; Hina et al. 2021). In 1997, the extremes of the country had the most dry conditions, followed by the interior, which had the most humid conditions. This contrasted with 1998, when the extremes of the country had the most dry conditions and the interior had the most humid conditions. There was a predominance of the category similar to normal, followed by fairly dry and occasionally severely humid categories in 2015. The very dry and extremely dry weather categories dominated in 2016, as they did in 1998. According to Chaudhry and Rasul (2000), Pakistan has around two-thirds of its land covered by semi-arid and arid climates. The results obtained also match the findings of Hanif et al. (2013), where they found that Southwest Pakistan is especially vulnerable to drought.

Temporal Pakistan's homogeneous rainfall groups are depicted in Fig. 8 using the SPI-12 temporal distribution. In the time series, SPI-12 oscillations revealed dry and wet seasons. In this study, we focused only on assessing drought in homogeneous populations. As a result of the G_1 group, Baldin and Karachi (Airport) stations are included, where

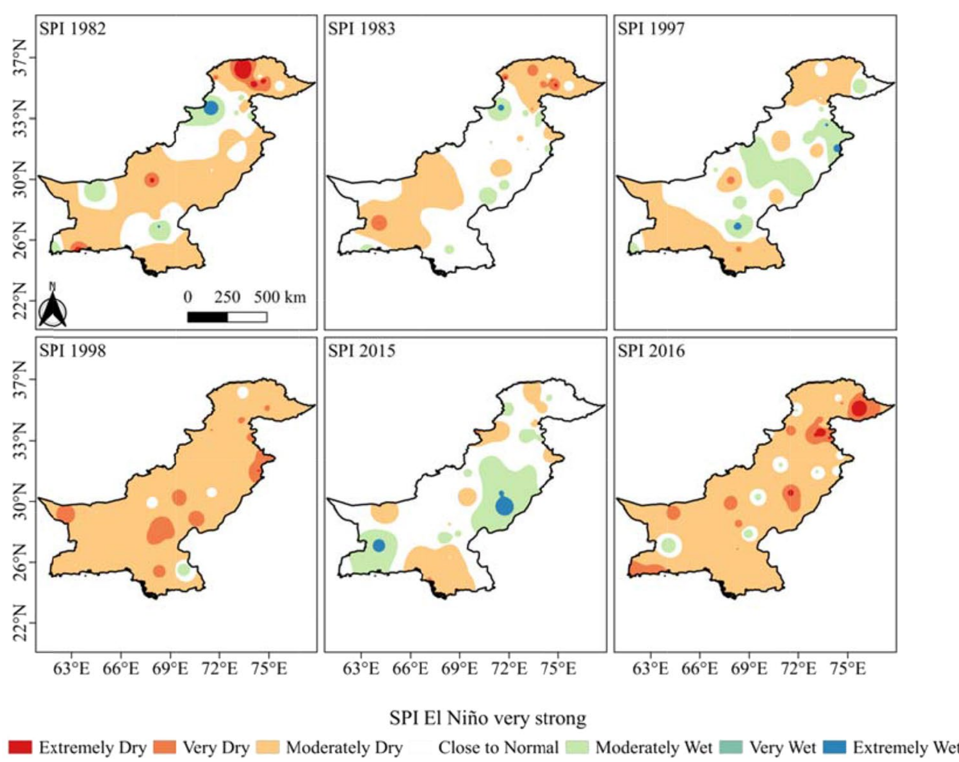

Fig. 7 Special distribution of SPI-12 in the years of El Niño events in the very strong category, according to Oceanic Niño Index (ONI) in Pakistan

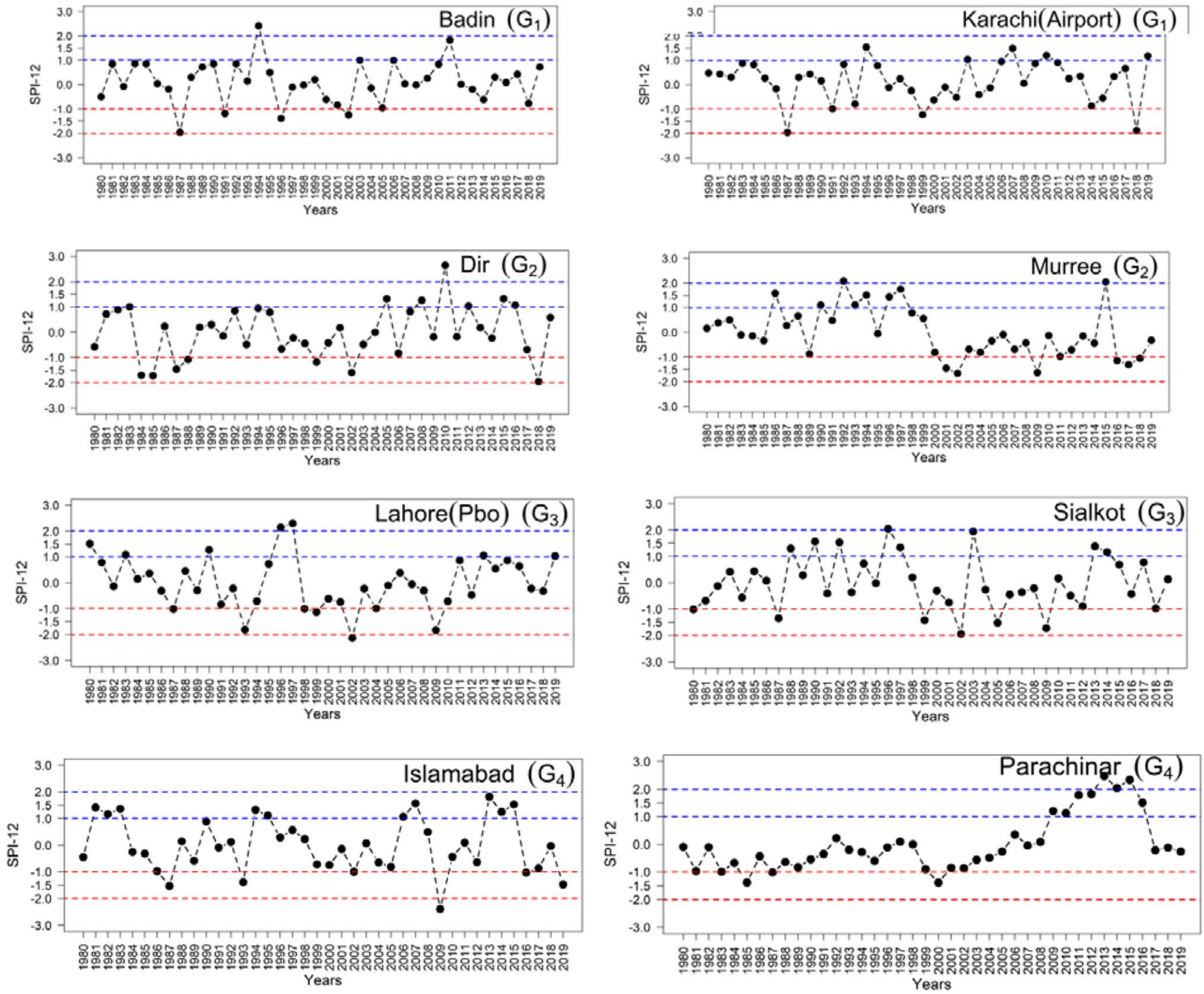

Fig. 8 SPI-12 temporal distribution of homogeneous rainfall groups (G_1, G_2, G_3, and G_4) during 1980–2019

the SPI-12 category is close to normal, followed by some years categorized as very dry (SPI-12-1.0), for example, the years 1989 (La Niña strong), 1991 (El Niña strong), 1996 (La Niña moderate), 2002 (El Niño moderate), and 2006 (La Niña weak), the exceptions of which were the years 1987 and 2018 (Table 3). As in G_1, the Dir and Murree stations fall into the G_2 group, which has a high prevalence of this category, but with a higher degree of variation among these stations in this category. Aside from a year classified as extremely dry in 2018 in Dir season, the other years were 1984/1985 (La Niña Weak), 1987/1988 (El Niño Strong), and 1999 (La Niña Strong) (Huang et al. 2017; NOAA/CPC 2021).

This group, which includes Lahore (PBO) and Sialkot stations, is similar to the previous groups, with the exception of 2002 (El Niño Moderate), which was categorized as extremely dry — SPI-12 — in the Lahore (PBO) station in the years 1987 (El Niño strong), 1993/94 (La Nia strong), 1999/2000 (La Niña strong), 2004/05 (El Niño weak), and 2009/2010 (El Niño moderate) (Table 3). Of all of the stations studied over time, only the years 1987 (El Niño strong) and 2009 (El Niño moderate) at the Sialkot station had significant drought occurrences (Huang et al. 2017; NOAA/CPC 2021). A similar pattern is seen in G_4 group, with the Islamabad and Parachinar stations, with the Islamabad station showing a predominance of the category near normal, followed by SPI-12's "extremely dry" category, except for year 2009, when the Parachinar station showed the majority of drought events in the 1980s, except for 1999/2000. According to Hina et al. (2021), 1987, 1991, 2000, 2001, 2002, and 2005 are the years in which the SPI and the Single *Z*-Index cross information in the temporal evaluation.

According to Khan and Gadiwala (2013), public managers should pay attention to drought episodes in the country

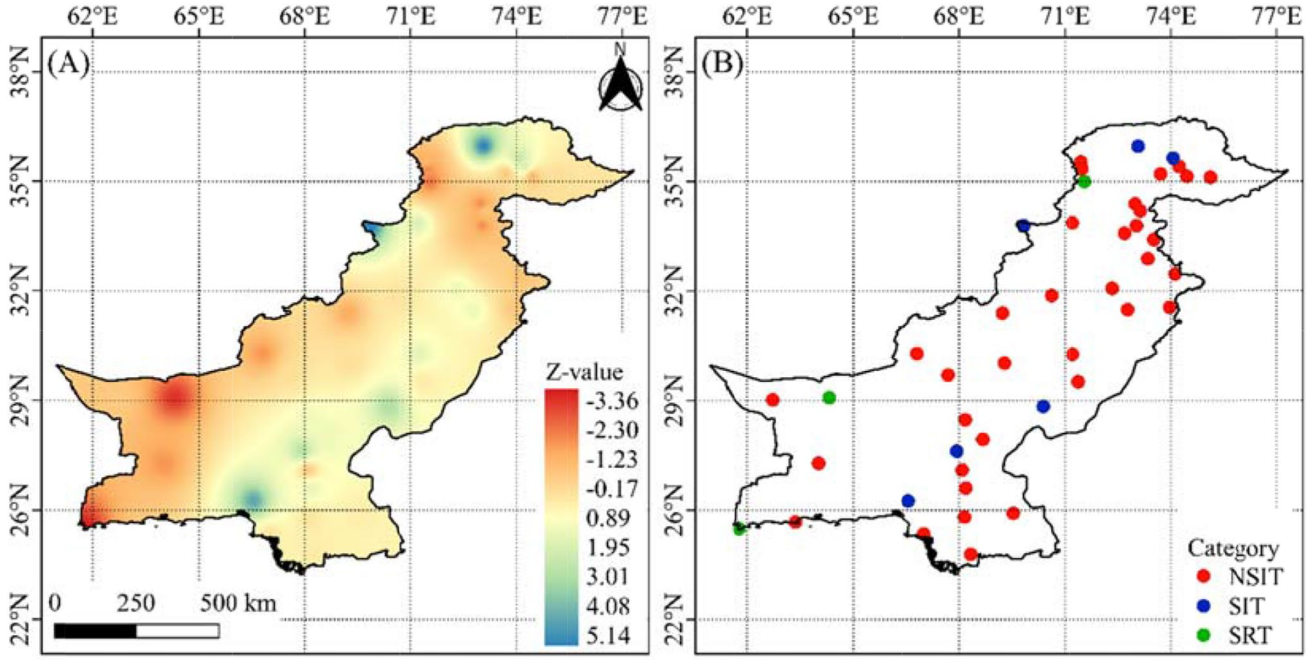

Fig. 9 Annual rainfall trend (mm) in Pakistan based on Mann-Kendall (MK) test results versus *Z* value (**A**) and its Z_{MK} categories (**B**) in the period 1980–2019

since the country has been struggling with water scarcity (Xie et al. 2013), which is exacerbated by drought and brings social and economic impacts and increased social vulnerability (Ahmed et al. 2018; Adnan et al. 2018).

3.4 Annual rainfall trend

Statistical analysis of rainfall trend (Fig. 9) via MK test (Z and Z_{MK}) showed that most of the stations used in the study were observed in the NSIT rainfall category (Table 2) throughout Pakistan, the exceptions were in the stations in the south (category SRT–Dir–ID 12, Jiwani–ID21, and Dalbandin–ID10) and in the North (category STI-Gilgit–ID15, Gupis–ID16, Khanpur (Pbo)–ID23, Lasbella–ID26, Moen-Jo-Daro–ID27 and Parachinar–ID35) (Table 5). The values obtained from $Z > 0$ and p-value > 0.05 of probability were only in the stations that were categorized as STI and SRT, on the contrary, in the rest of the country with the predominance of conditions of $Z < 0$ and p-value < 0.05 probability, mainly in the Balochistan Plateau (Fig. 1). Spatially, the annual rainfall reduction trend is concentrated in the north and south sectors, that is, obeying the previously identified rainfall gradient (Fig. 2), and conversely, increasing trends were observed in specific sectors existing in the country in the last 40 years. The results obtained agree with the results obtained by Salma et al. (2012), who showed a significant downward trend in rainfall; however, only in 20 years of the series, followed by the results of Hanif et al. (2013), which showed that annual rainfall significantly increased in Pakistan's high latitudes. Similarly, Praveen et al. (2020) identified 11 meteorological divisions that recorded a notable rainfall decline trend for the monsoon season in India from 1901 to 2015.

It is noteworthy that the MK test is the most appropriate method for locating and approximately detecting the starting point of a given trend, being the most suitable method for analyzing climate change in time series (Gois et al. 2020; Terassi et al. 2020). The non-significant increase in the magnitude of rainfall in Pakistan, according to Sen's method, was recorded mainly in homogeneous groups G_2, G_3, and G_4, the exception being group G_1 (Table 5).

3.5 CWT analysis

In a signal, the patterns that progress over time are identified while applying time-localized filtering that encourages to identify spectral characteristics. To categorize the interrelation of wavelet coefficient with every signal component, the wavelet maps are retrieved. With the periodicity and their spatial localization, the continuous wavelet power spectrum offered an approximate idea.

The analyses for annual SPI for 40 years over 8 cities were considered as a continuous series, where interesting patterns can be observed. For the purpose of long-term trend detection, the 40-year period (1980–2019) was subdivided

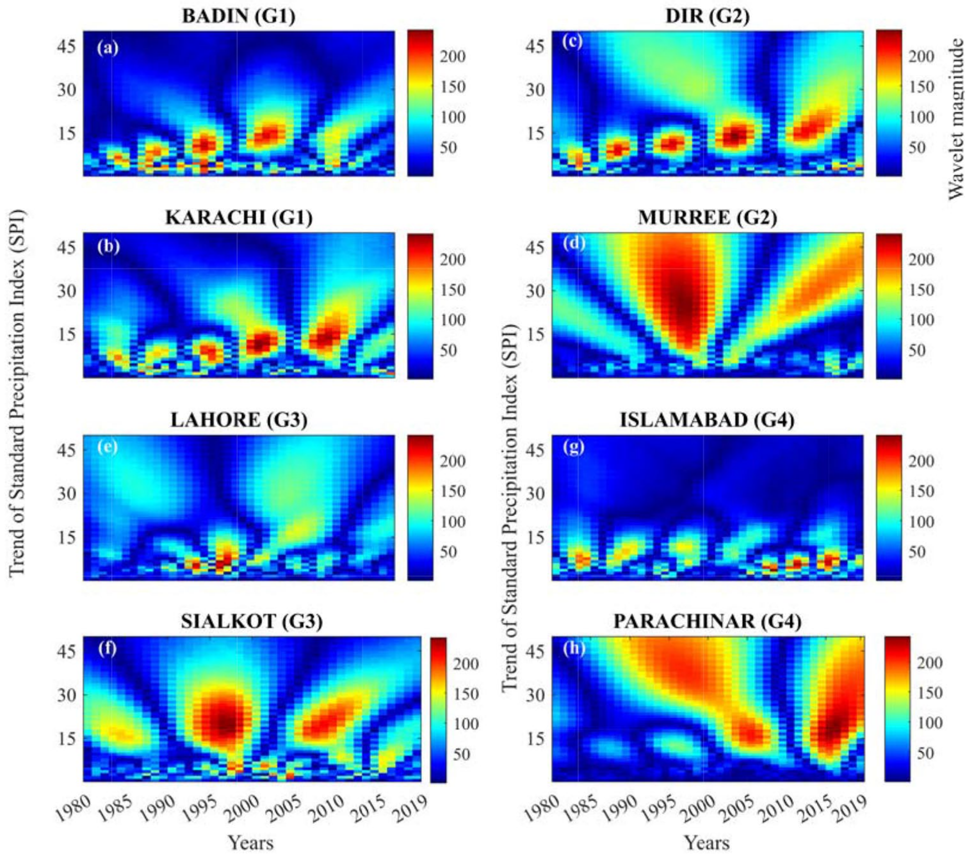

Fig. 10 Annual SPI trend in Pakistan from 1980 to 2019

into 4 groups i.e., G_1 (Badin, Karachi), G_2 (DIR, Murree), G_3 (Lahore, Sialkot), and G_4 (Islamabad, Parachinar). In G_1, the Karachi station retrieved SPI data showed positive behavior during first 5 years of analyzed data followed by less rainfall conditions in year 1987 (Fig. 10b), Badin station also showed similar pattern of SPI data. During the year 1991 and 1996, the drier weather conditions were classified overall, but in Karachi the SPI values were higher as compared to Badin (Fig. 10a).

In G_2, the Dir station shows dry years while the years 2005, 2010, and 2012 marked somewhat wet surroundings (Fig. 10c), wherein 2014 and 2015 showed extreme rainfall values that lead to reduced dry conditions in 2017 and extreme pattern during 2018. The apted station in Murree recorded moderate dry conditions roughly during 2005 to 2010, and the year 2015 showed extreme rainfall, i.e., adequate number of rainfall followed by dry conditions (negative SPI) for next 4 years (Fig. 10d).

The G_3 which includes Lahore and Sialkot showed dry conditions in 1993 accompanied by wet conditions during 1995 which then turns in to extreme wet for 1996 and 1997 (Fig. 10e, f). During year 2003m Lahore depicted moderately dry condition, i.e., negative SPI values, while Sialkot recorded positive SPI values, i.e., wet conditions. During the last years (2017–2019), we observed the transition from dry to wet conditions meaning by significant increase in rainfall level gradually. While considering G_4, we observed continuous dry conditions during the second decade of our considered data in Parachinar but in comparison to this behavior Islamabad station showed dispersed rainfall throughout the data (Fig. 10g, h). During 2010 and onwards, we also measured extreme rainfall values for Parachinar stations and Islamabad replicated scattered values of SPI.

4 Conclusions

Rainfall and drought in Pakistan are assessed through applied statistics. Based on statistical criteria, the Average, McQuitty and Centroid methods are applicable to historical rainfall series in Pakistan in defining homogeneous rainfall regions. Cluster analysis can identify four homogeneous rainfall regions in Pakistan (G_1, G_2, G_3 and G_4). Descriptive, multivariate and exploratory statistics showed that G_1 differs from the other groups in size and spatial distribution. Pettitt's technique identified the most extreme El Niño years in terms of spatial and temporal drought variability.

In Pakistan, the Pettitt test applied to the time series of monthly rainfall shows that March, August, September, June and December are the months with the greatest rainfall variation. These months should be monitored by public managers in Pakistan to the greatest extent possible.

SPI-12 categorizes the performance of the Strong and Very Strong El Niños in Pakistan into relatively dry and close to normal categories. SPI-12 once again reveals that Pakistan's rainy seasons are dominated by the categories "near to normal" and "very dry," but severely dry years occur in the country, and these years in particular demand attention by Pakistani managers.

It has been observed that the Mann-Kendall test shows non-significant increases in Pakistan's rainfall, with the exception of a few stations in the southern and northern regions of Pakistan. It is apparent that the non-significant increase in Pakistan's rainfall is of concern to numerous sectors of society, such as planning and management territory, agricultural, tourism, commerce and urban drainage and water resources management, as well as the economy, tourism and agriculture. The CWT also endorsed the findings of the statistical results by significant trend in different cluster groups. The G1 group showed significant trend in different year of the study period.

The recommendations for further studies are related to the application of Gaussian geostatistical modeling to rainfall and drought data at the monthly, quarterly and semiannual scales of the SPI, intercomparison between drought indices (SPEI, RDI, among others), influence of multiscale meteorological systems on the dynamics of rainfall and drought and assess future scenarios proposed by the Intergovernmental Panel on Climate Change (IPCC) for drought and rainfall in Pakistan, according to the latest technical report for 2021.

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Code availability Not applicable.

Declarations

Ethics approval There is no ethical conflict by all authors.

Consent to participate All authors agree to participate.

Consent for publication All authors agree with the publication.

Conflict of interest The authors declare no competing interests.

References

ADB (2017) Climate risk and vulnerability analysis report. Baluchistan water resources development Project preparatory technical assistance (TA 8800-PAK). Asian Development Bank, Government of Balochistan, JICA Japan and Techno-Consult International

Adnan S (2010) Spatio-temporal distribution of drought and its characteristics over Pakistan (Doctoral dissertation, COMSATS University Islamabad, Islamabad, Pakistan)

Adnan S, Ullah K, Shuanglin L, Gao S, Khan AH, Mahmood R (2018) Comparison of various drought indices to monitor drought status in Pakistan. Clim Dyn 51(1):1885–1899. https://doi.org/10.1007/s00382-017-3987-0

Ahmad I, Zhaobo S, Weitao D, Ambreen R (2010) Trend analysis of January temperature in Pakistan over the period of 1961-2006: Geographical perspective. Pak J Meteorol 7(1):11–22

Ahmad I, Ambreen R, Sultan S, Sun Z, Deng W (2014) Spatial-temporal variations in January precipitation over the period of 1950-2000 in Pakistan and possible links with global teleconnections: geographical perspective. Am J Clim Chang 3(4):378–387. https://doi.org/10.4236/ajcc.2014.34033

Ahmed K, Shahid S, Nawaz N (2018) Impacts of climate variability and change on seasonal drought characteristics of Pakistan. Atmos Res 214(1):364–374. https://doi.org/10.1016/j.atmosres.2018.08.020

Alsubih M, Mallick J, Talukdar S, Salam R, AlQadhi S, Fattah MA, Thanh NV (2021) An investigation of the short-term meteorological drought variability over Asir Region of Saudi Arabia. Theor Appl Climatol 145(1):597–617. https://doi.org/10.1007/s00704-021-03647-4

Anjum SA, Saleem MF, Cheema MA, Bilal MF, Khaliq T (2012) An assesment to vulnerability, extent, characteristics and severity of drought hazard in Pakistan. Pak J Meteorol 64(1):85–96

Beguería S, Vicente-Serrano SM, Reig F, Latorre B (2010) Standardized precipitation evapotranspiration index (SPEI) revisited: parameter fitting, evapotranspiration models, tools, datasets and drought monitoring. Int J Climatol 34(10):3001–3023. https://doi.org/10.1002/joc.3887

Brito TT, Oliveira-Júnior JF, Lyra GB, Gois G, Zeri M (2017) Multivariate analysis applied to monthly rainfall over Rio de Janeiro state, Brazil. Meteorog Atmos Phys 129(5):469–478. https://doi.org/10.1007/s00703-016-0481-x

Carmo MVNS, Lima CHR (2020) Caracterização Espaço-Temporal das Secas no Nordeste a partir da Análise do Índice SPI. Revista Brasileira de Meteorologia 35(2):233–242. https://doi.org/10.1590/0102-7786352016

Chaudhry QZ, Rasul G (2000) Agroclimatic classification of Pakistan, Science Vision, 9(1–2 & 3–4), (July–Dec, 2003 & Jan–Jun, 2004), 59

Costa MS, Oliveira-Júnior JF, Santos PJ, Correia Filho WLF, Gois G, Blanco CJC, Silva Junior CA, Santiago DB, Souza EO, Jardim AMRF (2021) Rainfall extremes and drought in Northeast Brazil and its relationship with El Niño–Southern Oscillation. Int J Climatol 41(S1):E2111–E2135. https://doi.org/10.1002/joc.6835

Eckstein D, Künzel V, Schäfer L, Winges M (2020) Global climate risk index. Germanwatch, Bonn, pp 1–43

Fung KF, Huang YF, Koo CH, Soh YW (2020) Drought forecasting: a review of modelling approaches 2007–2017. J Water Clim Chang 11(3):771–799. https://doi.org/10.2166/wcc.2019.236
Gois G, Oliveira-Júnior JF, Silva Junior CA, Sobral BS, Terassi PMB, Leonel Junior AHS (2020) Statistical normality and homogeneity of a 71-year rainfall dataset for the state of Rio de Janeiro—Brazil. Theor Appl Climatol 141(1):1573–1591. https://doi.org/10.1007/s00704-020-03270-9
Gonçalves STN, Vasconcelos Junior FC, Sakamoto MS, Silveira CS, Martins ESPR (2021) Índices e Metodologias de Monitoramento de Secas: Uma Revisão. Revista Brasileira de Meteorologia 36(3):495–511. https://doi.org/10.1590/0102-7786363000
Hafeez A, Ehsan M, Abbas A, Shah M, Shahzad R (2022) Machine learning-based thermal anomalies detection from MODIS LST associated with the Mw 7.7 Awaran, Pakistan earthquake. Nat Hazards. https://doi.org/10.1007/s11069-021-05131-8
Hanif M, Khan AH, Adnan S (2013) Latitudinal precipitation characteristics and trends in Pakistan. J Hydrol 492(7):266–272. https://doi.org/10.1016/j.jhydrol.2013.03.040
Heim RR Jr (2002) A review of twentieth-century drought indices used in the United States. Bull Am Meteorol Soc 83(8):1149–1166. https://doi.org/10.1175/1520-0477-83.8.1149
Hina S, Saleem F, Arshad A, Hina A, Ullah I (2021) Droughts over Pakistan: possible cycles, precursors and associated mechanisms. Geomat Nat Haz Risk 12(1):1638–1668. https://doi.org/10.1080/19475705.2021.1938703
Ho S, Tian L, Disse M, Tuo Y (2021) A new approach to quantify propagation time from meteorological to hydrological drought. J Hydrol B 603:127056. https://doi.org/10.1016/j.jhydrol.2021.127056
Huang B, Thorne PW, Banzon VF, Boyer T, Chepurin G, Lawrimore JH et al (2017) Extended reconstructed sea surface temperature, version 5 (ERSSTv5): upgrades, validations, and intercomparisons. J Clim 30(20):8179–8205. https://doi.org/10.1175/JCLI-D-16-0836.1
Hussain MS, Lee SH (2009) A classification of rainfall regions in Pakistan. J Kor Geogr Soc 44(5):605–623
Jardim AMRF, Silva MV, Silva AR, Santos A, Pandorfi H, Oliveira-Júnior JF, Lima JLMP, Souza LSB, Araújo Junior GN, Lopes PMO, Moura GBA, Silva TGF (2021) Spatiotemporal climatic analysis in Pernambuco State, Northeast Brazil. J Atmos Sol Terr Phys 223(1):105733. https://doi.org/10.1016/j.jastp.2021.105733
Karim R, Tan G, Ayugi B, Babaousmail H, Liu F (2020) Evaluation of historical CMIP6 model simulations of seasonal mean temperature over Pakistan during 1970–2014. Atmosphere 11(9):1005. https://doi.org/10.3390/atmos11091005
Kendall MG (1975) Rank correlation measures. Charles Griffin, London, p 220
Khan MA, Gadiwala MS (2013) A study of drought over Sindh (Pakistan) using Standardized Precipitation Index (SPI) 1951 to 2010. Pak J Meteorol 9:15–22
Kim CJ, Park MJ, Lee JH (2014) Analysis of climate change impacts on the spatial and frequency patterns of drought using a potential drought hazard mapping approach. Int J Climatol 34(1):61–80. https://doi.org/10.1002/joc.3666
Lyra GB, Oliveira-Júnior JF, Zeri M (2014) Cluster analysis applied to the spatial and temporal variability of monthly rainfall in Alagoas state, Northeast of Brazil. Int J Climatol 34(13):3546–3558. https://doi.org/10.1002/joc.3926
Lyra GB, Oliveira-Júnior JF, Gois G, Cunha-Zeri G, Zeri M (2017) Rainfall variability over Alagoas under the influences of SST anomalies. Meteorog Atmos Phys 129(1):157–171. https://doi.org/10.1007/s00703-016-0461-1
Mallick J, Talukdar S, Almesfer MK, Alsubih M, Ahmed M, Reza A, Islam T (2022) Identification of rainfall homogenous regions in Saudi Arabia for experimenting and improving trend detection techniques. Environ Sci Pollut Res 29(1):25112–25137. https://doi.org/10.1007/s11356-021-17609-w
Mann HB (1945) Non-parametric test against trend. Econometrica 13(1):245–259. https://doi.org/10.2307/1907187
McKee TB, Doesken NJ, Kleist J (1993) The relationship of drought frequency and duration to time scales. In: Proceedings of the 8th Conference on Applied Climatology, 17, 179-183
Mishra AK, Singh VP (2010) A review of drought concepts. J Hydrol 391(1-2):202–216. https://doi.org/10.1016/j.jhydrol.2010.07.012
Modarres R, Sarhadi A (2011) Statistically-based regionalization of rainfall climates of Iran. Glob Planet Chang 75(1):67–75. https://doi.org/10.1016/j.gloplacha.2010.10.009
Mohsenipour M, Shahid S, Chung E, Wang X (2018) Changing pattern of droughts during cropping seasons of Bangladesh. Water Resour Manag 32(1):1555–1568. https://doi.org/10.1007/s11269-017-1890-4
NOAA/CPC - National Oceanic and Atmospheric Administration/Climate Prediction Center (2021) Cold & Warm Episodes by Season. [online]. Available at: http://www.cpc.ncep.noaa.gov/products/analysis_monitoring/ensostuff/ensoyears.shtml. Accessed 21 November 2021
Oliveira Júnior JF, Lyra GB, Gois G, Brito TT, Moura NDSHD (2012) Análise de homogeneidade de séries pluviométricas para determinação do índice de seca IPP no estado de Alagoas. Floresta e Ambiente 19(1):101–112. https://doi.org/10.4322/floram.2012.011
Oliveira-Júnior JF, Xavier FMG, Teodoro PE, Gois G, Delgado RC (2017) Cluster analysis identified rainfall homogeneous regions in Tocantins state, Brazil. Biosci J (On line) 33(1):333–340. https://doi.org/10.14393/BJ-v33n2-32739
Oliveira-Júnior JF, Gois G, Lima Silva IJ, Oliveira Souza E, Jardim AMRF, Silva MV, Jamjareegulgarn P (2021) Wet and dry periods in the state of Alagoas (Northeast Brazil) via Standardized Precipitation Index. J Atmos Sol Terr Phys 224(1):105746. https://doi.org/10.1016/j.jastp.2021.105746
Pettitt AN (1979) A non-parametric approach to the change-point problem. J R Stat Soc: Ser C: Appl Stat 28(1):126–135. https://doi.org/10.2307/2346729
Praveen B, Talukdar S, Shahfahad MS, Mondal J, Sharma P, Reza A, Islam MT, Rahman A (2020) Analyzing trend and forecasting of rainfall changes in India using non-parametrical and machine learning approaches. Sci Rep 10(1):10342. https://doi.org/10.1038/s41598-020-67228-7
R Core Team (2020) R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL https://www.R-project.org/
Rashid MM, Beecham S, Chowdhury RK (2015) Assessment of trends in point rainfall using continuous wavelet Transforms. Adv Water Resour 82(1):1–15. https://doi.org/10.1016/j.advwatres.2015.04.006
Salma S, Rehman S, Shah MA (2012) Rainfall trends in different climate zones of Pakistan. Pak J Meteorol 9(17):37–47
Sayama T, Ozawa G, Kawakami T, Nabesaka S, Fukami K (2012) Rainfall–runoff–inundation analysis of the 2010 Pakistan flood in the Kabul River basin. Hydrol Sci J 57(2):298–312. https://doi.org/10.1080/02626667.2011.644245
Silva MV, Pandorfi H, Jardim AMRF, Oliveira-Júnior JF, Divincula JS, Giongo PR, Silva TGF, Almeida GLP, Moura GBA, Lopes PMO (2021) Spatial modeling of rainfall patterns and groundwater on the coast of Northeastern Brazil. Urban Clim 38(1):100911. https://doi.org/10.1016/j.uclim.2021.100911
Slingo JM, Annamalai H (2000) 1997: The El Niño of the Century and the response of the Indian summer monsoon. Mon Weather Rev 128(6):1778–1797. https://doi.org/10.1175/1520-0493(2000)128<1778:TENOOT>2.0.CO;2

Sobral BS, Oliveira Júnior JF, Gois G, Pereira Junior ER (2018) Spatial variability of SPI and RDI drought indices applied to intense episodes of drought occurred in Rio de Janeiro State, Brazil. Int J Climatol 38(10):3896–3916. https://doi.org/10.1002/joc.5542

Sokal RR, Rohlf FJ (1962) The comparison of dendrograms by objective methods. Taxon 11(2):33–40. https://doi.org/10.2307/1217208

Terassi PMB, Oliveira Júnior JF, Gois G, Galvani E, Sobral BS, Biffi VHR (2020) Analysis of daily rainfall and spatiotemporal trends of extreme rainfall at Paraná slope of the Itararé watershed, Brazil. Revista Brasileira de Meteorologia 35(2):357–374. https://doi.org/10.1590/0102-7786352025

Ullah I, Ma X, Yin J, Asfaw TG, Azam K, Syred S, Liu M, Arsahd M, Shahzaman M (2021) Evaluating the meteorological drought characteristics over Pakistan using in situ observations and reanalysis products. Int J Climatol 41(9):4437–4459. https://doi.org/10.1002/joc.7063

Wilks DS (2011) Statistical methods in the atmospheric sciences, 3rd edn. Academic Press, Oxford, p 704

WMO – World Meteorological Organization (2012) Standardized precipitation index user guide. Available in: http://www.wamis.org/agm/pubs/SPI/WMO_1090_EN.pdf. Access on: 24 Nov. 2021

Xie H, Ringler C, Zhu T, Waquas A (2013) Droughts in Pakistan: a spatiotemporal variability analysis using the Standardized Precipitation Index. Water Int 38(5):620–631. https://doi.org/10.1080/02508060.2013.827889

Yerdelen C, Abdelkader M, Eris E (2021) Assessment of drought in SPI series using continuous wavelet analysis for Gediz Basin, Turkey. Atmos Res 260(1):105687. https://doi.org/10.1016/j.atmosres.2021.105687

Zargar A, Sadiq R, Naser B, Khan FI (2011) A review of drought indices. Environ Rev 19(1):333–349. https://doi.org/10.1139/a11-013

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